CE / CS / SE 3354 · SOFTWARE ENGINEERING · FALL 2026

Project Deliverable P2

Analysis and High-Level Design — Version 1.0

Weight 6% · Due Sunday 8 November 2026, 11:59 PM Central · One PDF on Canvas, the same PDF committed to /docs/p2 and tagged p2

Field	Your entry
Project title	QR Code Restaurant Menu & Ordering System (QRMOS)
Team number and section	Team 01 · Section 003
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Repository URL	https://github.com/mrzetoon/team-01-QRRestaurantOrder
Document version	1.0 (grown from the P1 baseline 0.1)
Date	8 November 2026

0 Revision History and Contribution Record

Version history, who produced what for P2 (with repository evidence), and every change made since the P1 baseline, including our answer to each point of the P1 feedback.

Version	Date	Author	Summary of change
0.1	2026-10-18	All members	P1 baseline (submitted on Canvas; committed as /docs/p1/P1-submitted.pdf)
0.5	2026-11-01	Mohmmad Abuzetoon, Rafay Alvi	P1 feedback answered (0.2); change requests CR-01 to CR-06 applied to sections 2 and 3
0.8	2026-11-04	Eshitasri Punuganti, Xuan Mai Tran, Len Isherwood	Sections 4, 5 and 8: expanded use cases, analysis models, spike 01 result; CR-07 to CR-10
0.9	2026-11-07	Isaiah Bueno-Hernandez, Eshitasri Punuganti	Sections 6, 7 and 9: high-level design, decision records, traceability; cross-checks K1 to K6 run
1.0	2026-11-08	All members	P2 baseline: reviewed, signed (section 12), committed to /docs/p2 and tagged p2

0.1 Individual contribution record

Team member	What this person produced for P2	Evidence in the repository
Mohmmad Abuzetoon	Section 0 (records and the answers to P1 feedback in 0.2); sections 10 to 12; repository restructure to the 10.4 layout; one issue per section; assembled the PDF, committed it to /docs/p2 and pushed tag p2.	Commits by @mrzetoon: 5aa7ace; PR #2; review of PR #5
Rafay Alvi	Sections 1 to 3 under change control: CR-01 to CR-06, FR-18 to FR-20, FR-14 source fix, use case diagram replacement in 3.4.	Commits by @MuhammadAlvi03: 82b1bb3, b3f610a; PR #5
Eshitasri Punuganti	Section 4: UC-01, UC-03 and UC-07 expanded turn by turn, and the self-review log; section 9: traceability matrix and gap analysis.	Commits by @peshi24: d9ca5f9; PR #6; review of PR #7 github branch: eshiupload
Xuan Mai Tran	Section 5: domain model, analysis object list, scenario tables and both analysis sequence diagrams (sources in /docs/p2/diagrams); ran K1 to K6 and kept the defect log.	Commits by @keasark: 1298185; PR #3, #4; commented on and merged PR #2 Github branch: member04_xuanmai
Isaiah Bueno-Hernandez	Sections 6 and 7: quality-attribute scenarios, style choice, component and deployment views, mapping, design principles, ADR-01 to ADR-05; filed the restaurant manager interview notes in /docs/elicitation.	Commits by @buenoisaiah006-bit: cb8ef58 PR # 7; reviewer of PR # 6
Len Isherwood	Section 8: spike 01 (live order delivery) — server, benchmark, 12 measured runs and README in /spikes/spike-01-live-orders.	No P2 pull request before the deadline (only P1 work, merged in PR #1)

0.2 Changes since the P1 baseline

Every point of the P1 feedback, what we changed, and where. Requirement changes were raised as change requests (CR) below and approved by team vote before ADR-01 chose the architecture.

P1 feedback point	What we changed	Where
C1 — every stakeholder need was still marked as assumed; the restaurant interview was planned but not done	**Isaiah Bueno-Hernandez interviewed** the manager of Arabeska Restaurant, Plano, on 8 Nov 2026 . The manager confirmed the needs of S1–S4 in 2.2 and answered Q1–Q6, Q8 and Q9. Five answers contradicted our assumptions and became change requests CR-11 to CR-17 before the design was revised.	2.2, 11, CR-11–CR-17, /docs/elicitation/2026-11-08-restaurant-manager-interview.md
C1 — the submitted version replaced the cost estimate with an open range ($8,000–$75,000)	Stakeholder S6 now cites the itemized estimate of $43,263 (2.6). The range appears only as the budget ceiling in C3 and CON-03.	2.2, 2.6, CR-02
C4 — /docs folders, meeting notes and issues described in 4.4 were missing	Repository rebuilt to the layout in 10.4: /docs/p1, /docs/p2, /docs/p2/diagrams, /docs/elicitation, /docs/meetings, /spikes. Meeting notes for every weekly meeting since 18 Oct; one issue per P2 section.	10.4, repository
C4 — only three members had commits	Each member owns a P2 section and committed it from their own account on a branch, merged through a pull request that another member reviewed.	0.1
C4 — the PDF in the repository differed from the one on Canvas	The Canvas P1 PDF is committed as /docs/p1/P1-submitted.pdf. For P2 the same file goes to /docs/p2 and is tagged p2 before one member uploads it.	10.4
Leftover placeholders (an unfilled pull-request number and an unfilled member name)	Removed. Section 0.1 cites only commits and pull requests that exist.	0.1
Repository URL on the cover page	Corrected and checked by opening it while logged out.	Cover
FR-14 cited S5, which is not a source of that need	FR-14 now cites S2, S3 and S4 (the staff and manager who log in). Q7 no longer names the instructor as a funder.	3.1, 11, CR-01
Record the interview results and update requirements under change control before choosing an architecture	CR-01 to CR-10 were approved before ADR-01 (4 Nov). The interview raised CR-11 to CR-17, which changed UC-01, UC-03, UC-07, the domain model and ADR-05 (8 Nov).	0.2, 4–7, 9

Change requests approved since the P1 baseline

ID	Change	Reason	Affects
CR-01	FR-14 source changed from S4, S5 to S2, S3, S4.	P1 feedback	3.1
CR-02	S6 need cites the itemized estimate instead of a range.	P1 feedback	2.2
CR-03	Use case diagram replaced: the "Payment process" oval ("we won't build it") is gone; UC-07 Pay Bill with the Payment Gateway as supporting actor; «include» Browse Menu and Log In; «extend» Customize Item and Cancel Order.	The P1 diagram contradicted FR-16, FR-17 and the 3.5 matrix (found by K3)	3.4
CR-04	FR-18 added: a resubmitted cart is recorded only once.	UC-01 alternate 13a (section 4.2)	3.1
CR-05	FR-19 added: the staff queue shows time since each order was received.	UC-03 step (0) (section 4.3)	3.1
CR-06	FR-20 added: the customer sees a payment receipt.	UC-07 acknowledgment, step (8) (section 4.4)	3.1
CR-07	Risk R2 fallback changed from polling every 3 s to every 1 s.	Spike 01 measured 3-s polling below the QR-01 target	10.3, ADR-02
CR-08	Expanded use cases moved from 3.4 to section 4 and rewritten turn by turn; 3.4 keeps the diagram and one-line scopes.	P2 structure	3.4, 4
CR-09	UC-01 precondition "the menu has at least one available item" became alternate flow 3a.	Self-review question 6	4.2
CR-10	P1 risk R5 mitigation corrected: it described load time, not labour hours.	Internal review	10.3
CR-11	FR-21 added: all orders of a table session are grouped on one table check. Replaces our assumption of separate tickets.	Interview Q3 (Q2)	3.1, 4.3, 5.1

ID	Change	Reason	Affects
CR-12	FR-22 added: staff can cancel a single order line with a reason. UC-03 alternate changed from whole-order cancel.	Interview Q4 (Q3)	3.1, 4.3, 5.1
CR-13	FR-23 added: a server confirms each new order before the kitchen sees it (new status Awaiting confirmation).	Interview Q5 (Q4)	1.3, 3.1, 4.2, 4.3, 5.3
CR-14	FR-24 added: menu in English and Spanish.	Interview Q6 (Q5)	3.1, 5.1
CR-15	FR-25 added: each item lists its common allergens (was free text in the description).	Interview Q7 (Q6)	3.1, 5.1
CR-16	FR-26 added: staff see a clear alert when a customer's payment fails.	Interview Q8 (Q8)	3.1, 4.4
CR-17	FR-27 added: phones at one table share one cart. ADR-05 reversed: the cart is kept on the server.	Interview Q9 (Q9)	3.1, 6.5, 7.5

1 Introduction

1.1 Purpose of this document and of the system

This document is version 1.0 of the requirements and design baseline for the QR Code Restaurant Menu & Ordering System (QRMOS). It grows the P1 baseline (version 0.1) under change control: sections 1 to 3 are carried forward with requirement IDs preserved, and sections 4 to 9 add use-case analysis, analysis models and the high-level design that P3 will implement. It follows ISO/IEC/IEEE 29148 [R1] and is written for the team, the CS 3354 instructor, and the restaurant manager deciding whether to adopt the system.

QRMOS addresses the wait and the errors in ordering at a sit-down restaurant. A diner currently waits for a server to bring a menu, waits again to place the order, and depends on the server copying it correctly onto a ticket for the kitchen; at busy times this is where most delay and wrong orders come from. Printed menus also cannot show that an item has sold out or that a price has changed. QRMOS lets a diner scan a QR code on the table, read the current menu on their own phone, customize items, send the order straight to the staff tagged with the table number, and pay the bill through a payment gateway. Staff see orders arrive in real time and update their status; the manager keeps the menu current.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
A unique QR code per table that opens the menu for that table	Storing or processing card numbers ourselves — card entry happens only on the payment gateway's hosted page, which keeps card data out of our system [R5]
Digital menu by category with description, price, photo and availability	Integration with the restaurant's POS or accounting system — vendor APIs differ and need credentials we cannot get
Item customization (options, quantity, special note) and a cart	Takeout, delivery and table reservations — off-premises ordering is a different problem with different actors
Order submission tagged with the table number, plus order status for the diner	Customer accounts, loyalty points and reviews — not needed to order, and they add personal data we would have to protect
Live staff order queue with status updates (Received → Preparing → Ready → Served)	Native iOS/Android apps — a mobile web page needs no install, which is the point of scanning a code
Menu management, sold-out flags, table setup, and staff login with Manager/Staff roles	Multi-restaurant (franchise) management, inventory tracking and sales analytics
Paying the table's bill online through an external payment gateway (hosted checkout)	Splitting a bill between diners, tips and refunds — staff handle these at the till as today

Why the exclusions were excluded: each excluded item either brings legal or security obligations (storing card data, customer accounts) or depends on systems we do not control (POS), so including it would put the MUST requirements at risk inside one semester. None of them is needed for the core value: a diner can order from the table and the kitchen receives it correctly.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
QRMOS	QR Code Restaurant Menu & Ordering System — the system specified here.
QR code	Quick Response code: a 2-D barcode [R4] printed on a table card that a phone camera turns into a web link (URL).
QR-nn	**Not** a QR code: the identifier of a Quality Requirement in section 3.2 (template convention). Recorded here because the two meanings collide in this project.
Table session	The period from a diner opening a table's link until they close it or it expires; all orders in it carry that table number.
Order	One submission of a cart from one table, with its own order number. All orders of a table session are grouped on one table check (CR-11, interview [R9]).
Order status	Where an order is in its life: Awaiting confirmation (until a server confirms it, CR-13), Received, Preparing, Ready, Served, or Cancelled.
Modifier / option	A choice attached to a menu item, e.g. size, spice level, add extra cheese (+$1.00).
Sold out	An item that is shown on the menu but cannot be ordered right now. Different from **removed**, which means the item no longer appears at all.
Staff / Manager	Staff = server or kitchen employee who processes orders. Manager = owner or shift lead who can also edit the menu, tables and accounts.
Real time	In this document: visible on the other screen within 3 seconds without a page reload (QR-01).
POS	Point of sale: the restaurant's existing till / card terminal. Outside QRMOS.
Payment gateway	An outside service (e.g. Stripe, Square) that shows its own card-entry page, charges the card and tells QRMOS approved or declined. A supporting actor in UC-07.
«include» / «extend»	Use case links: «include» = a step the base use case always uses (Log In, Browse Menu); «extend» = optional behaviour added only in some cases (Customize Item, Cancel Order).
FR, CON, UC, MoSCoW	Functional requirement; constraint requirement; use case; priority scale MUST / SHOULD / COULD / WON'T.
Analysis object	An object found while analysing a use case (e.g. TableMenu, OrderBook). Named after the problem, never after a technology (section 5).
TUCBW / TUCEW / TUCCW	This use case begins with / ends with / continues with — where an expanded use case starts, stops, or hands over.
QAS	Quality-attribute scenario: stimulus, environment, response and a measurable response measure (section 6.1).
ADR	Architecture decision record: one decision, the alternatives, and what it costs us (section 7).
Spike	A small, time-boxed program built only to answer one design question with a measurement (section 8).
Trust boundary	A line in the deployment view where data passes between zones we trust differently (section 6.4).

1.4 References

[R1] ISO/IEC/IEEE 29148:2018, Systems and software engineering — Life cycle processes — Requirements engineering.

[R2] CS 3354 Fall 2026 syllabus, P1 template and P2 project description; course textbook Chapters 4–8.

[R3] W3C, Web Content Accessibility Guidelines (WCAG) 2.1, Level AA.

[R4] ISO/IEC 18004:2015, QR Code bar code symbology specification.

[R5] PCI Security Standards Council, PCI DSS v4.0 (cited to justify keeping card entry on the gateway's page, CON-02).

[R6] Team brainstorm notes, 10 Oct 2026, /docs/elicitation/2026-10-10-brainstorm.md.

[R7] S. E. Şeker, CS 3354 Lessons 13–14, Use Case Analysis and Object Interaction Modelling (expanded use cases, scenario tables, checks K1–K6).

[R8] Spike 01 README and raw results, /spikes/spike-01-live-orders/ (section 8).

[R9] Restaurant manager interview, 8 Nov 2026, **by Isaiah Bueno-Hernandez** ; notes in /docs/elicitation/2026-11-08-restaurant-manager-interview.md.

2 Overall Description

2.1 Product perspective and context

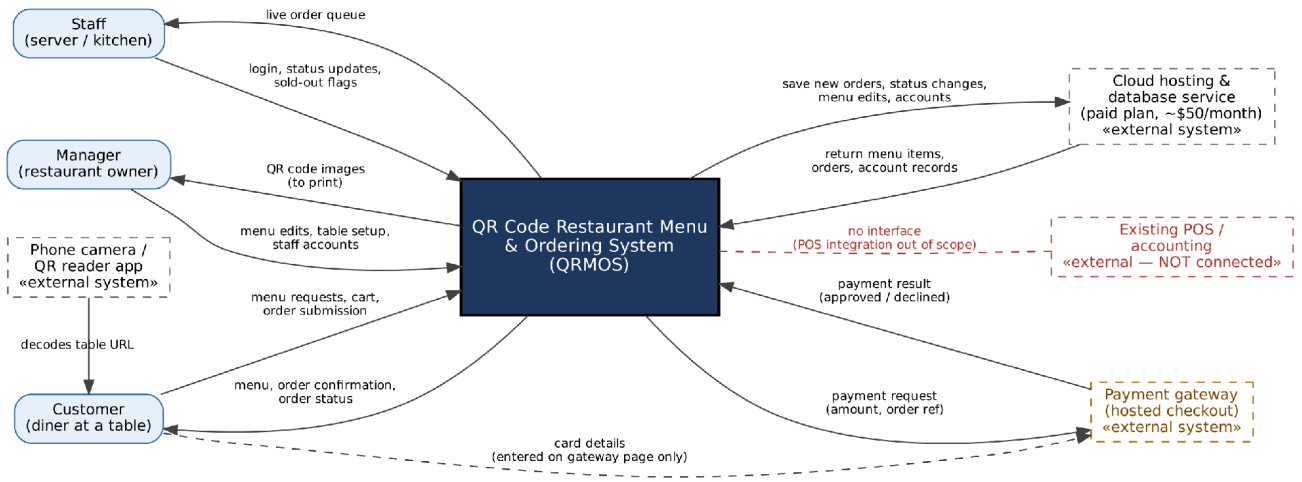


Figure 1. Context diagram. Everything inside the dark box is ours to build: the customer menu pages, the staff order queue, the manager pages, and the logic and data behind them. The boundary sits at the phone browser on one side and the hosting/database service on the other: we use the phone camera, the cloud host and the payment gateway but build none of them, so they are drawn as external systems. The customer types card details only on the gateway's own page; QRMOS sends the amount and receives approved or declined, so card data never enters our box. The existing POS is drawn on purpose with **no** interface, which makes that exclusion visible instead of silent.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
S1 Customer (diner)	Primary user; scans the code and orders	Order without waiting; see real prices and what is sold out; know the order was received	Confirmed by the manager interview [R9]: customers mostly want fast service; sold-out items and wrong orders happen at busy

Stakeholder	Role in relation to the system	What they need from it	How you learned this
			times. Reported by the manager, not asked of diners directly.
S2 Server	User; delivers food, handles problems at the table	Fewer trips to take orders; orders clearly tagged by table	Confirmed in part by the manager [R9]: wants a server to confirm each order before the kitchen (CR-13) and a clear alert when a payment fails (CR-16). Servers not interviewed directly.
S3 Kitchen staff	User; prepares food from the order queue	Legible orders with every option and note, in arrival order	Confirmed in part by the manager [R9]: the kitchen cancels single items, not whole orders (CR-12). Kitchen staff not interviewed directly.
S4 Manager / owner	Pays for and operates the system; edits menu	Change menu and sold-out items instantly; faster table turnover; a build cost within budget and running cost under about $400/month	Confirmed by the manager interview [R9]: uses Square, restaurant pays the fee; wants one check per table (CR-11), English and Spanish (CR-14), listed allergens (CR-15) and one shared cart per table (CR-17).
S5 CS 3354 instructor & TAs	**Never uses it**; assesses the project	Testable, traceable requirements; repository access; on-time milestones	Read from the syllabus and P1 template [R2]
S6 Development team	Builds and maintains it this term	Scope that fits the term; the itemized estimate of $43,263 (2.6), inside the $75,000 ceiling (C3)	Agreed at the weekly team meeting (Saturdays, 2:00 pm, Microsoft Teams); minutes in /docs/meetings
S7 Payment gateway provider	**Never uses the screens**; external service that processes card payments (supporting actor in UC-07)	Requests in its API format, a merchant account for the restaurant, its per-transaction fee	Assumed from providers' public developer documentation; which provider is open question Q1

2.3 User characteristics

Customers. Any adult or teenager in the restaurant, from tech-comfortable students to older diners who rarely scan QR codes. They use their own phone (any size, iOS or Android, any modern browser), often one-handed, in dim light and in a hurry to order before the rest of the table. They cannot be assumed to read English fluently, to have good eyesight, or to know restaurant terms. Some will use screen readers or large text, so the menu must work with both (QR-04). They will not create an account or install anything.

Staff (servers and kitchen). Know the menu and the restaurant's routine very well but have no training in this system and no time to read instructions during service. They will look at a shared tablet or screen from a distance, sometimes with wet or gloved hands, so the order queue needs large text and large buttons. Turnover is high, so a new staff member must be productive after a short demo.

Manager. Comfortable with a laptop and with editing a spreadsheet-like list, but not a programmer. Edits the menu occasionally, usually before opening, and sets up tables once. Needs to print QR codes on an ordinary office printer.

2.4 Constraints

C1 — Web application, no install. Customers must reach the menu through a phone web browser; no native app. Source: project definition agreed by the team and stated in our proposal.

C2 — Time. The system must be specified, designed, built and tested within the Fall 2026 term, by the P3 deadline. Source: syllabus [R2].

C3 — Budget. Total cost for building the system, its hardware and the first year of hosting must stay between $8,000 and $75,000. Our estimate is about $43,300 (see 2.6). Source: CS 3354 project instructions [R2].

C4 — No card data in our system. Payment must go through an external payment gateway's hosted page; QRMOS must never collect, transmit or store card numbers. Source: scope decision in 1.2 and PCI DSS [R5].

C5 — Instructor access. All source code and documents must live in a repository the instructor can open. Source: P1 template section 4.4 [R2].

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	Most diners carry a smartphone with a camera, a browser and mobile data or Wi-Fi.	Diners without a phone cannot order; the restaurant must keep paper menus and server ordering as a fallback, and adoption numbers fall.
A2	The restaurant has a reliable internet connection during service.	Orders are delayed or lost and the staff queue shows stale data; QR-01 cannot be met.
A3	The restaurant has at least one tablet or PC that stays open on the staff order screen.	No one sees incoming orders, so UC-03 has no actor at the moment orders arrive.
A4	(Dependency) A paid cloud hosting and database plan (about $50/month) stays available and priced as quoted for the first year.	We lose the demo environment and must migrate mid-project, threatening C2 and the P3 demo.
A5	(Dependency) The restaurant's Square merchant account (interview [R9]) offers a hosted checkout page we can send customers to.	Online payment (UC-07) cannot go live; diners pay the server as today, and FR-16/FR-17 slip to a later release (see Q1).
A6	A QR code on a table is only scanned by people sitting at that table.	Someone could photograph a code and send fake orders from outside; we would need staff confirmation of orders (see Q4).

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Feasible	Every piece is a common web feature. Spike 01 measured live order delivery on our planned design: push reached the staff screen in 0.3 s at the 95th percentile with 150 ms network delay, well inside QR-01 (section 8).
Operational	Feasible, with one risk	It copies the existing workflow (order → server check → kitchen → serve) and only moves where the order is typed. The manager interview [R9] confirmed the problems it targets and asked for a server confirmation step (CR-13), which keeps staff in control. Risk: older diners may not want to scan; A1 fallback keeps paper menus.
Schedule	Feasible if scope holds	MUST items are 9 of 27 functional requirements (33%); the interview added seven SHOULD items (CR-11 to CR-17), which can be dropped first if work runs late. The plan to P3 in 10.2 has dated owners; risk R3 (scope creep) is handled by change control (0.2).
Economic / resource	Feasible	Estimated total about $43,300, within the $8,000–$75,000 budget (C3): • Development labour: 6 people × 10 weeks × 10 h = 600 h at an assumed $60/h = $36,000 • Cloud hosting and database, paid plan: $50/month × 12 = $600 • Domain name: $20/year • Payment gateway: no monthly fee assumed; the restaurant pays a per-transaction fee (about 3% per payment, to confirm in Q1) • Two staff tablets for the order screen: 2 × $350 = $700 • Printed, laminated QR table cards: 20 tables × $5 = $100 • Usability-test participant gift cards: 10 × $20 = $200 • Contingency 15% = $5,643 → total $43,263 Running cost after year 1: about $4,200/year (hosting $600, domain $20, maintenance about 10% of build $3,600). Rates are estimates to confirm (Q7).

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-01	The system shall generate a QR code for each registered table that encodes a URL unique to that table.	MUST	S4	Register tables 1–3, generate codes, scan each with a phone: each opens the menu showing its own table number, and no two codes decode to the same URL.
FR-02	The system shall display the menu, grouped by category, to anyone who opens a valid table URL without asking them to log in or install an app.	MUST	S1, C1	Open a table URL in a fresh private browser on iOS Safari and Android Chrome: the menu appears and no login or install prompt is shown.
FR-03	The system shall show each menu item's name, description, price and availability.	MUST	S1, S4	Load a 20-item test menu: every item on screen matches the database record in all four fields.
FR-04	The system shall prevent a customer from adding a sold-out item to the cart.	SHOULD	S1, S3	Mark one item sold out: its Add button is disabled, and a direct add request to the server is rejected.
FR-05	The system shall let a customer choose an item's defined options, quantity and a special note of up to 200 characters before adding it to the cart.	MUST	S1, S3	Add a burger with size Large, quantity 2 and a 200-character note: the cart shows all three; a 201st character cannot be typed.
FR-06	The system shall let a customer change the quantity of, or remove, any cart item before submission, recalculating the total after each change.	SHOULD	S1	Change 2 → 3 and remove one item: the displayed total equals the hand-calculated total after each step.
FR-07	The system shall submit the cart as one order tagged with the table number of the URL it was opened from.	MUST	S2, S3	Submit from table 5's URL: the order in the database and on the staff screen shows table 5.
FR-08	The system shall show the customer a confirmation with an order number after a successful submission.	SHOULD	S1	After submitting, the confirmation number on the phone matches the order number on the staff screen.
FR-09	The system shall add each new order to the staff order queue, in submission-time order, without the staff member reloading the page.	MUST	S2, S3	With the staff screen open, submit 3 orders from 3 phones: all 3 appear in the right order with no reload (timing tested in QR-01).
FR-10	The system shall let a staff member move an order's status from Received to Preparing to Ready to Served, or to Cancelled.	MUST	S2, S3	Step one order through every status and cancel another: each change is saved with a timestamp; skipping or reversing a status is refused.
FR-11	The system shall show the customer the current status of each order submitted from their device in the current table session.	SHOULD	S1	Change an order's status on the staff screen: the customer's page shows the new status within 3 seconds without a reload.
FR-12	The system shall let a manager add, edit and remove menu items, including name, description, price, category, options and photo.	MUST	S4	Add, edit and delete one item: each change appears on a freshly loaded customer menu.
FR-13	The system shall let a staff member or manager mark a menu item sold out or available again.	SHOULD	S2, S3, S4	Mark an item sold out on the staff screen: a freshly loaded customer menu shows it as sold out; mark it available and it can be added again.

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-14	The system shall require a staff or manager account to log in with a username and password before any staff or manager page is shown.	MUST	S2, S3, S4 (CR-01)	Open every staff/manager page in a logged-out browser: each redirects to the login page.
FR-15	The system shall let a manager add, rename and deactivate tables, and a deactivated table's URL shall stop accepting orders.	COULD	S4 (found via 3.5)	Deactivate table 4 and scan its code: the page says the table is not active and the Submit action is unavailable.
FR-16	The system shall let a customer pay all unpaid orders of their table session through the external payment gateway's hosted payment page.	SHOULD	S1, S4	Submit two orders from table 3 and choose Pay: the gateway page shows the combined total; a test card payment succeeds.
FR-17	The system shall mark the paid orders as Paid only after the payment gateway confirms the payment as approved.	SHOULD	S4	Pay with a test card the gateway declines: orders stay Unpaid and the customer is told; pay with an approved test card: orders show Paid on the staff screen.
FR-18	The system shall record a cart only once when the customer resubmits it after a failed submission.	SHOULD	S1, S3 (CR-04)	Cut the connection just after Submit, then retry twice: the staff queue shows exactly one order for that cart.
FR-19	The system shall show, for each active order in the staff queue, the time elapsed since it was received.	SHOULD	S2, S3 (CR-05)	Submit an order and wait 5 minutes: the staff queue shows 5 min for it, within ±1 min.
FR-20	The system shall show the customer a receipt with the amount, time and payment reference after an approved payment.	SHOULD	S1 (CR-06)	Pay with an approved test card: the receipt shows the amount paid, the time and the gateway's reference.
FR-21	The system shall group all orders of a table session on one table check, shown to staff and used for the bill.	SHOULD	S4 (CR-11)	Submit two orders from table 4: the staff screen shows one check for table 4 listing both, and the bill totals both.
FR-22	The system shall let a staff member cancel a single order line with a reason, recalculating the order total.	SHOULD	S3 (CR-12)	Cancel one of three lines with reason "86'd": the other two stay, the total drops by that line's price, and the reason is recorded.
FR-23	The system shall show each new order to servers as Awaiting confirmation and send it to the kitchen only after a server confirms it.	SHOULD	S2, S4 (CR-13)	Submit an order: the kitchen view does not show it until a server confirms; after confirming, it appears as Received.
FR-24	The system shall show the customer menu in English or Spanish, as the customer chooses.	SHOULD	S1, S4 (CR-14)	Switch the language to Spanish: every category, item name and description shown is the Spanish text entered by the manager.
FR-25	The system shall show each menu item's common allergens.	SHOULD	S1, S4 (CR-15)	Tag an item with milk and nuts: both allergens appear on the item for the customer.
FR-26	The system shall alert staff with the table and amount when a customer's payment fails.	SHOULD	S2, S4 (CR-16)	Pay with a test card the gateway declines: the staff screen shows an alert with the table number and amount within 3 s.
FR-27	The system shall let every phone at the same table add to and see one shared cart.	SHOULD	S1, S4 (CR-17)	Open table 2 on two phones and add one item on each: both phones show both items and the same total.

3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	**Performance.** 95% of submitted orders shall appear on an open staff order screen within 3 seconds of submission, under 20 tables ordering at the same time.	MUST	S2, S3	Scripted test sends 100 orders from 20 simulated clients; compare submit and display timestamps; pass if ≥ 95 are ≤ 3 s.
QR-02	**Performance.** The customer menu shall show its first category of items within 3 seconds of opening a table URL in 95% of loads on a mid-range phone over a 4G connection.	MUST	S1	20 loads with Chrome DevTools "Fast 4G" throttling and a mid-range phone profile; record time to first item.
QR-03	**Usability.** At least 8 of 10 first-time users shall place an order of 2 items, one with an option chosen, in under 2 minutes without help.	SHOULD	S1	Moderated test with 10 students outside the team; timed from scan to confirmation screen; results in /docs/usability.
QR-04	**Accessibility.** The customer menu, cart and confirmation pages shall meet WCAG 2.1 Level AA [R3].	SHOULD	S1, R3	Zero serious/critical issues reported by axe DevTools, plus a full order completed using VoiceOver and TalkBack.
QR-05	**Security.** 100% of staff and manager server endpoints shall reject requests without a valid session, and manager-only endpoints shall reject Staff-role sessions.	MUST	S4	Automated test calls every endpoint logged-out, as Staff and as Manager; pass only if every expected 401/403 is returned.
QR-06	**Availability.** The system shall be reachable at least 99% of the time during a 7-day trial from 11:00 to 22:00 daily.	COULD	S4	An uptime monitor checks the menu URL every 5 minutes; uptime % computed from its log.

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The customer interface shall run in the phone's web browser without installing any app.	C1	It is the defining idea of the product; a diner will not install an app for one meal.
CON-02	The system shall not collect, transmit or store card numbers; card entry shall happen only on the payment gateway's hosted page.	C4, R5	Handling card data ourselves triggers full PCI DSS compliance the team cannot meet.
CON-03	The total cost of building the system, its hardware and the first year of hosting shall not exceed $75,000.	C3, R2	It is the budget ceiling set for the project; anything above it cannot be funded.
CON-04	All source code and documents shall be kept in a repository the instructor can access.	C5, R2	Required by the course for grading and contribution evidence.

3.4 Initial use cases

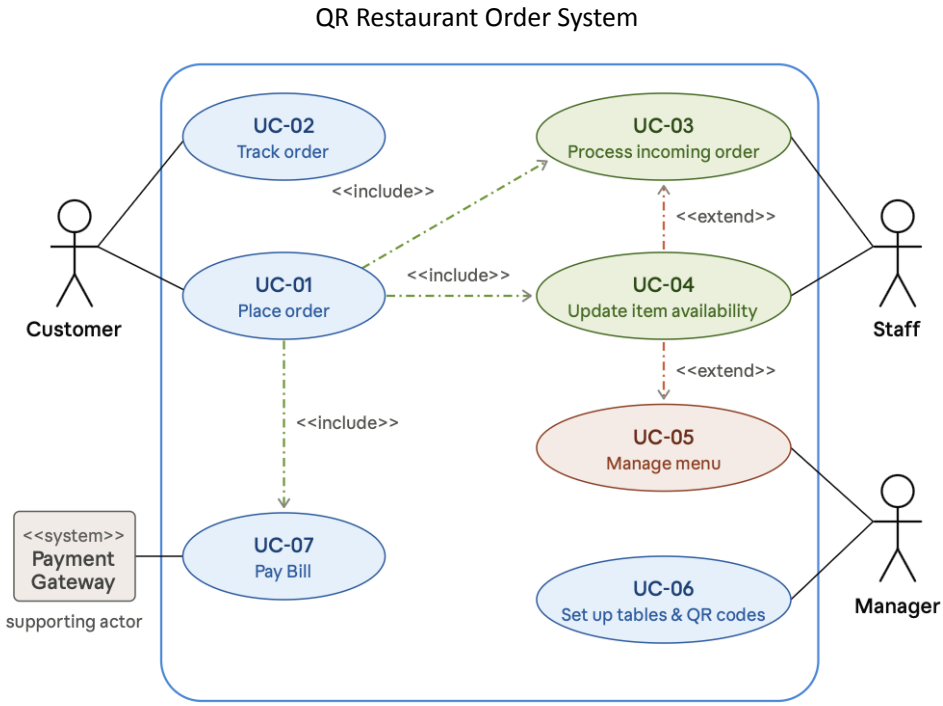


Figure 2. Use case diagram (revised by CR-03). Seven actor-goal use cases: UC-01, UC-02 and UC-07 for Customer, UC-03 and UC-04 for Staff, and UC-05 and UC-06 for Manager. «include»: Place Order always includes UC-03 Process Incoming Order, UC-04 Update Item Availability and UC-07 Pay Bill. «extend»: UC-04 extends UC-03 and UC-05. The Payment Gateway is a supporting actor of UC-07: it never starts a use case. Manager is a specialised Staff actor, so a manager can also process orders and mark items sold out.

High-level use cases (expanded turn by turn in section 4 where marked)

ID	Use case · actor	Begins with (TUCBW)	Ends with (TUCEW)	Level
UC-01	Place Order · Customer	the customer scans the QR code on the table	the customer reads the order confirmation	Expanded, 4.2
UC-02	Track Order Status · Customer	the customer opens the status of their orders	the customer reads the current status	High-level
UC-03	Process Incoming Orders · Staff	a new order appears in the staff queue	the staff member sees the order leave the active queue	Expanded, 4.3
UC-04	Update Item Availability · Staff	a staff member marks an item sold out or available	the staff member sees the new availability	High-level
UC-05	Manage Menu · Manager	the manager opens the menu editor	the manager sees the saved change	High-level
UC-06	Set Up Tables & QR Codes · Manager	the manager adds, renames or deactivates a table	the manager sees the table list and its printable code	High-level
UC-07	Pay Bill · Customer (Payment Gateway supports)	the customer chooses to pay	the customer reads the receipt	Expanded, 4.4

3.5 Requirements to use case traceability

Requirement ID	Use case(s) that realise it	If blank — why, and what you will do
FR-01	UC-06 (used by UC-01)	
FR-02	UC-01	
FR-03	UC-01	
FR-04	UC-01, UC-04	
FR-05	UC-01	
FR-06	UC-01	
FR-07	UC-01	
FR-08	UC-01	
FR-09	UC-03	
FR-10	UC-03	
FR-11	UC-02	Only requirement behind UC-02; if FR-11 were dropped, UC-02 would go with it.
FR-12	UC-05	
FR-13	UC-04	
FR-14	Log In «include» of UC-03, UC-04, UC-05, UC-06	
FR-15	UC-06, UC-01 (alt 2a)	Added in this revision — see below.
FR-16	UC-07	
FR-17	UC-07	
FR-18	UC-01 (alt 13a)	Added by CR-04.
FR-19	UC-03 (step 0)	Added by CR-05.
FR-20	UC-07 (step 7)	Added by CR-06.
FR-21	UC-03, UC-07	Added by CR-11.
FR-22	UC-03 (alt 6a)	Added by CR-12.
FR-23	UC-03 (steps 2–3)	Added by CR-13.
FR-24	UC-01 (step 2)	Added by CR-14.
FR-25	UC-01 (step 4)	Added by CR-15.
FR-26	UC-07 (alt 6a)	Added by CR-16.
FR-27	UC-01 (steps 8–10)	Added by CR-17.
QR-01 … QR-06	—	Non-functional: traced to components and decisions in section 9 instead.

What the matrix revealed, and what we changed because of it:

1. **UC-06 had nothing to trace to except FR-01.** FR-01 says "each registered table", but no requirement let anyone register a table. We added FR-15 (add, rename, deactivate tables) and the alternative flow 2a in UC-01.

2. **UC-02 rests on one SHOULD requirement (FR-11).** We kept it, but recorded that dropping FR-11 removes the whole use case.

3. **Role restriction first appeared as a functional requirement.** It reached no use case of its own, which showed it is a security quality, so it moved to QR-05.

4. **Ordering again from the same table** (a second round of drinks) has no requirement saying how it relates to the first order. Rather than invent an answer we raised it as Q2.

5. **Online payment had no use case.** Adding FR-16 and FR-17 needed a new goal, UC-07 Pay Bill, with the payment gateway as a supporting actor; CON-02 keeps card data on the gateway's page.

6. **Login was repeated in four use cases.** We made it one «include» (Log In) shared by UC-03 to UC-06, and split optional behaviour into «extend» parts (Customize Item, Cancel Order).

4 Use-Case Analysis

4.1 Which use cases we expanded, and why

Use case	Why it earns an expanded description
UC-01 Place Order	Carries the most requirements (FR-01 to FR-08, FR-18, FR-24, FR-25, FR-27) and every customer quality requirement; specified at P1.
UC-03 Process Incoming Orders	Where orders from many tables meet one staff screen (QR-01, QAS-1); changed most by the interview (CR-11 to CR-13); specified at P1.
UC-07 Pay Bill	The only use case with an outside party and with money at stake (FR-16, FR-17, CON-02); the riskiest to get wrong.

UC-02 and UC-04 to UC-06 have one obvious path each and stay at the high level in 3.4.

4.2 UC-01 Place Order

Field	Content
Primary / supporting actors	Customer / none (the order reaches staff through UC-03)
Preconditions (may be assumed)	The table is registered and its QR card is on the table. The customer's phone can open web links.
TUCBW · TUCEW	The customer scans the QR code on the table · the customer reads the order confirmation.
TUCCW	UC-02 Track Order Status, if the customer follows the order; UC-07 Pay Bill, if they choose to pay.
Postcondition	An order exists with a new number, the table number, every line, the total, the time and status Awaiting confirmation; it is on the table check and last in the staff queue; the customer has seen its number.
Invariant	The total shown always equals the sum of the line prices at the current menu prices.
Requirements realised	FR-01 to FR-08, FR-15 (2a), FR-18 (13a), FR-23, FR-24, FR-25, FR-27; QR-01 to QR-04

Customer · input and action	QRMOS · system response
	(0) The table card shows the table number and a QR code with "Scan to see the menu and order".
(1) The customer scans the QR code.	

Customer · input and action	QRMOS · system response
	(2) The system shows the menu for that table in the chosen language (English or Spanish): restaurant name, table number, categories, each item's name, price and whether it is sold out, and the table's shared cart (Browse Menu «include»).
(3) The customer selects an item.	
	(4) The system shows the item's description, photo, allergens, price, its options with their price changes, quantity 1 and an empty note.
(5) The customer chooses options, a quantity and, optionally, a note (Customize Item «extend», only when the item has options).	
	(6) The system shows the line price for those choices.
(7) The customer adds the line to the cart.	
	(8) The system shows the number of lines in the table's cart and its total, including lines added from other phones at the table. Steps 3–8 repeat until the customer has everything.
(9) The customer opens the cart.	
	(10) The system lists every line with its options, note, quantity and line price, and the total.
(11) The customer submits the order.	
	(12) The system checks that every item is still available and that the table is still taking orders.
	(13) The system records the order on the table's check with the next order number, status Awaiting confirmation and the time, and places it last in the staff queue.
	(14) The system shows the confirmation: order number, lines, total and status Awaiting confirmation.
(15) The customer reads the confirmation. (acknowledgment)	

Alternate	Condition	What happens
2a	The table is deactivated.	The system says the table is not taking orders and to ask a server; no ordering is possible. Ends. (FR-15)
3a	The item is sold out.	It is shown as sold out and cannot be selected; the customer picks another item. (FR-04)
5a	The note passes 200 characters.	The system accepts no more characters. (FR-05)
10a	The customer changes a quantity or removes a line.	The system recalculates the total and shows the cart again; if it is empty, submitting is not offered. (FR-06)
12a	An item sold out after it was added.	The system names the item, removes that line and shows the cart again; the customer may submit the rest. (FR-04)
12b	The table was deactivated after the scan.	As 2a; the cart is kept so a server can take it. (FR-15)
13a	The order cannot reach the restaurant.	The system says "Order not sent — try again" and keeps the cart; a retry records the order only once. (FR-18)
*a	The customer leaves without submitting.	The cart is discarded when the table session expires; no order exists.

4.3 UC-03 Process Incoming Orders

Field	Content
Primary / supporting actors	Staff member (a server confirms; kitchen staff prepare; a manager may act) / Customer, who sees the new status through UC-02
Preconditions (may be assumed)	The staff member is logged in (Log In «include», FR-14). The staff queue is open on the restaurant tablet (A3).
TUCBW · TUCEW	A new order appears in the staff queue · the staff member sees the order leave the active queue.
TUCCW	UC-04 Update Item Availability, after alternate 6a.
Postcondition	The order is Served (or Cancelled with a reason), and every status change is recorded with time and staff member.
Invariant	An order's status only moves forward (Awaiting confirmation → Received → Preparing → Ready → Served) or to Cancelled; the kitchen never sees an unconfirmed order.
Requirements realised	FR-09, FR-10, FR-13 (6a), FR-14, FR-19, FR-21, FR-22, FR-23; QR-01, QR-05

Staff member · input and action	QRMOS · system response
	(0) The staff queue shows every active order oldest first, grouped by table check, each with table number, lines with options and notes, status, and time since received.
	(1) The system adds the new order to its table's check, marked Awaiting confirmation, without the staff member reloading.
(2) A server reads the order and confirms it.	
	(3) The system records the confirmation with the time and the server, marks the order Received and shows it to the kitchen.
(4) A kitchen staff member marks the order Preparing.	
	(5) The system records the change with the time and the staff member, and shows the order as Preparing.
(6) The kitchen staff member marks the order Ready.	
	(7) The system records the change and shows the order as Ready.
(8) The server marks the order Served.	
	(9) The system records the change and moves the order from the active queue to the table's completed list.
(10) The staff member sees the queue without that order. (acknowledgment)	

Alternate	Condition	What happens
2a	The server finds the order wrong or fake (no one at the table) — Cancel Order «extend».	The server cancels the order with a reason; the kitchen never sees it; the customer is told. (FR-23, A6)
6a	The kitchen cannot make one item — Cancel Order «extend».	The staff member cancels that line with a reason; the system recalculates the order total, keeps the other lines and tells the customer. TUCCW UC-04 to mark the item sold out. (FR-22)
4b	The staff member tries to skip or reverse a status.	The system refuses and keeps the current status. (FR-10)
1a	The staff tablet has lost its connection.	The system shows that the queue is offline; when the connection returns it shows every order received meanwhile (QAS-2).
*a	The staff member's session has expired.	The system asks them to log in again and then shows the same queue.

4.4 UC-07 Pay Bill

Field	Content
Primary / supporting actors	Customer / Payment Gateway (outside our system)
Preconditions (may be assumed)	The restaurant has a merchant account with the payment gateway (A5).
TUCBW · TUCEW	The customer chooses to pay · the customer reads the receipt.
Postcondition	An approved payment with the gateway's reference is recorded against the bill, and every order on the bill is Paid.
Invariant	No order is ever Paid without an approved result from the gateway (FR-17); card details never pass through QRMOS (CON-02).
Requirements realised	FR-16, FR-17, FR-20, FR-21, FR-26; CON-02; QR-05

Customer · input and action	QRMOS · system response
	(0) The order status page lists the customer's orders for this table session with each status, which are unpaid, and the option to pay.
(1) The customer chooses to pay.	
	(2) The system shows the bill: every unpaid order of this table session with its total, and the amount due.
(3) The customer confirms the amount.	
	(4) The system passes the amount and the bill reference to the payment gateway and sends the customer to the gateway's checkout.
(5) The customer enters card details on the gateway's own page and approves. (outside our system)	
	(6) The system receives the approved result, records the payment against the bill and marks the bill's orders Paid.
	(7) The system shows the receipt: amount paid, time, gateway reference and the orders covered.
(8) The customer reads the receipt. (acknowledgment)	

Alternate	Condition	What happens
2a	Nothing is unpaid in this table session.	The system says there is nothing to pay. Ends.
5a	The customer cancels on the gateway's page.	The customer returns to the bill; the orders stay unpaid. Ends.
6a	The gateway declines the payment.	The system says the payment was declined and to try again or ask a server; staff see an alert with the table and amount; orders stay unpaid. (FR-17, FR-26)
6b	No result arrives within 2 minutes.	The system says it could not confirm the payment and to ask a server; staff see the same alert as 6a; orders stay unpaid until a result arrives. (FR-26)

4.5 Self-review log

Each use case was checked against the ten review questions [R7] by a member who did not write it: UC-01 by Isaiah Bueno-Hernandez, UC-03 by Xuan Mai Tran, UC-07 by Rafay Alvi.

#	Review question	UC-01	UC-03	UC-07
1	Actor is a role, not a person or "user"	Pass	Pass	Pass
2	There is a step (0)	Fixed — P1 had none	Fixed — P1 had none	Pass
3	Every row holds one turn	Fixed — P1 step 9 split into 12–13	Fixed — P1 step 5 split; confirm step added (CR-13)	Pass
4	Every actor-supplied item is a domain attribute	Fixed — note had no home (D1)	Fixed — cancel reason (D2)	Pass
5	Ends with a perceivable acknowledgment	Pass	Fixed — P1 ended silently	Fixed — receipt added (FR-20)
6	Preconditions are assumptions, not failures	Fixed — CR-09	Pass	Pass
7	Postcondition can become a test	Pass	Pass	Pass
8	Alternates named by their step	Pass	Pass	Pass
9	No background processing	Pass	Fixed — customer-view update removed	Pass
10	No technology, store or widget	Fixed — "button", "database" removed	Fixed — "Offline banner"	Fixed — "Stripe" removed

5 Analysis Models

5.1 Conceptual domain model

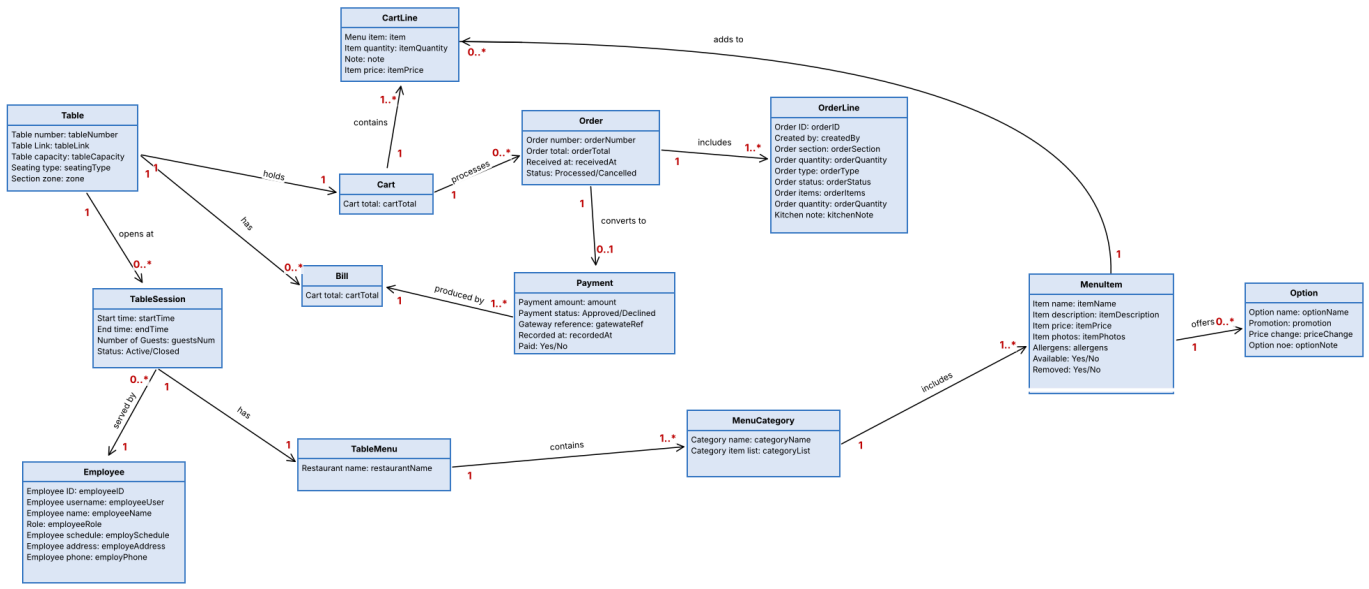


Figure 3. Conceptual domain model (analysis), revised after the interview. Thirteen concepts with attributes and multiplicities, no operations and no technology. Diners are anonymous, so there is no Customer concept: they reach the menu through the table's QR link. The payment gateway is outside our boundary and is not modelled: we keep only what it tells us (Payment.paymentStatus and gatewayRef). A Table holds one Cart, so all phones at the table share it, and each submitted Cart is processed into an Order made of OrderLines. A Bill is produced from the Payments made against it. A declined attempt is still a Payment, which is why a Bill has 1..* payments.

Item an actor supplies or reads	Use case step	Home in the domain model
table link (from the QR code)	UC-01 (1)	Table.tableLink
item, options, quantity, note	UC-01 (3), (5)	CartLine.item, itemQuantity, note (options from MenuItem *offers* Option) → OrderLine.orderItems, orderQuantity, kitchenNote
order number, status, total	UC-01 (14)	Order.orderNumber, status, orderTotal
status chosen, cancel reason	UC-03 (2)–(6), 2a	Order.status, OrderLine.orderStatus (cancel reason: no attribute yet, see gaps)
time since received	UC-03 (0)	Order.receivedAt (elapsed time is derived)
amount due, orders covered	UC-07 (2), (3)	Bill.cartTotal (to rename amountDue); orders reached through Table → Cart → Order
receipt: amount, time, reference	UC-07 (7)	Payment.amount, recordedAt, gatewayRef
card details	UC-07 (5)	None. Entered on the gateway's page only (CON-02)

5.2 Analysis objects

Object	What it is for (one line)	Inside?	Domain concept
Table	A physical table with its number, capacity, zone and QR link.	Inside	Table
TableSession	One sitting at a table, with start and end time, guest count and status.	Inside	TableSession
TableMenu	The restaurant's menu as shown to a table's session.	Inside	TableMenu
MenuCategory	Groups menu items under a category name.	Inside	MenuCategory
MenuItem	Knows its price, allergens, availability and its options.	Inside	MenuItem
Option	A choice on a menu item, with the price change it causes.	Inside	Option
Cart	Holds the lines chosen at a table and their total, until submitted.	Inside	Cart
CartLine	One chosen item with its quantity, note and price.	Inside	CartLine
Order	A submitted cart, with its number, total, received time and status.	Inside	Order

OrderLine	One line of an order, with its status and kitchen note for the staff.	Inside	OrderLine
Bill	The total owed for a table, settled by one or more payments.	Inside	Bill
Payment	One payment attempt and its result.	Inside	Payment
Employee	The staff member who serves a table session.	Inside (the record; the person is an actor outside)	Employee
PaymentGateway	Takes the card payment and reports the result.	Outside	— (supporting actor)

5.3 UC-01 Place Order — scenario table and analysis sequence diagram

Trivial steps (no row): (0) the card is physical; (26) the customer reads. **Non-trivial:** (1)–(7) fetch the menu and an empty cart; (8)–(13) show the menu, choose an item and price it with its options; (14)–(17) add the line and recalculate the cart; (18)–(24) process the order, create order and order lines, notify the employee, and confirm to the customer.

Step	Subject	Subject action	Other data / objects	Object acted upon
1	Customer	accesses the TableMenu (opens the table link via QR code)	QR link	TableMenu
2	TableMenu	retrieves the restaurant's menu	restaurant name	MenuCategory
3	TableMenu	requests the menu categories	—	MenuCategory
4	MenuCategory	returns the list of categories	categories	TableMenu
5	TableMenu	requests the items of a selected category (with prices and availability)	category name	MenuItem
6	MenuItem	returns the options of an item and their price changes	item	Option
7	TableMenu	retrieves the table's cart (initially empty)	—	Cart
8	TableMenu	displays the menu and the empty cart to the customer	menu, cart	Customer
9	Customer	chooses options, quantity, and a note for an item	item, options, quantity, note	TableMenu
10	TableMenu	requests the item's base price with the chosen options	options, quantity	MenuItem
11	MenuItem	returns each chosen option's price change	chosen options	Option
12	TableMenu	displays the line price to the customer	line price	Customer
13	Customer	adds the line to the cart	—	TableMenu
14	TableMenu	adds the line (item, options, quantity, note, line price) to the cart	item, options, quantity, note, line price	Cart
15	Cart	creates a cart line for the chosen item	item, quantity, note, item price	CartLine
16	Cart	recalculates the total	lines	Cart (itself)
17	TableMenu	displays the number of lines and the total to the customer	total	Customer
18	Customer	submits the order	—	TableMenu
19	TableMenu	checks whether the session is still active	—	(internal check)
20	TableMenu	processes the cart into an order	cart lines, total	Cart
21	Cart	creates the order with its number, total, received time, and status	order total, receivedAt, status	Order
22	Order	creates one order line for each cart line	item, quantity, kitchen note	OrderLine
23	TableMenu	notifies the employee serving the table of the new order	order number	Employee
24	TableMenu	displays the order number, status, and total to the customer	order number, status, total	Customer

UC-01 Place Order Sequence Diagram

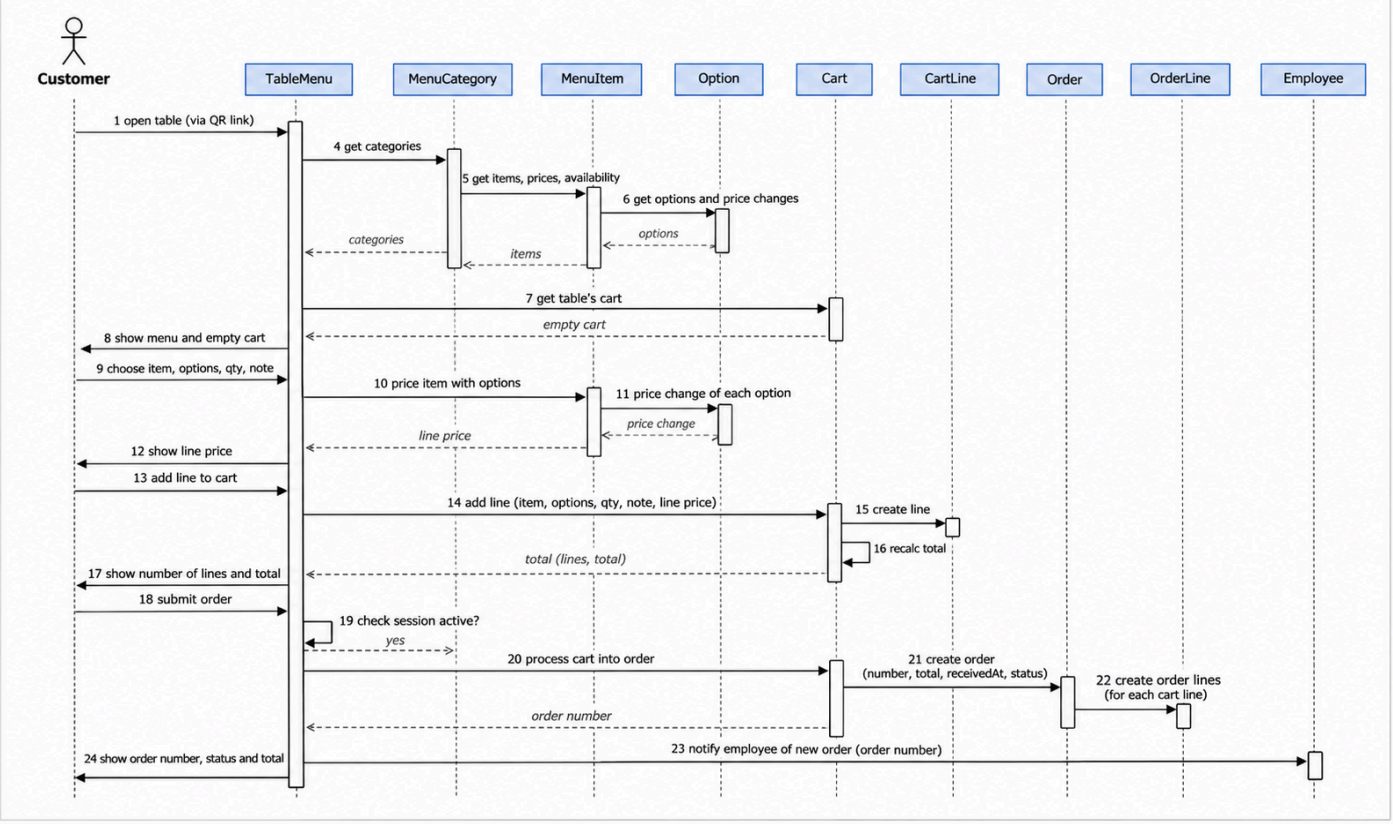


Figure 4. UC-01 analysis sequence diagram (revised). Messages 1–24 correspond to the revised scenario table rows one for one; each message name is a shortened form of the corresponding action. The Customer interacts directly with TableMenu, eliminating the Table and TableSession objects from the sequence diagram. TableMenu controls the main ordering process from message 1 through message 24, while returns are represented by dashed, unnumbered arrows. TableMenu retrieves the menu categories, items, options, and empty cart (4–7), displays the menu to the Customer (8), and handles item selection and pricing (9–12). The Customer adds the selected item to the cart (13), and TableMenu forwards the request to Cart (14). Cart creates a CartLine (15) and recalculates its total through a self-call (16, case D). When the Customer submits the order (18), TableMenu performs an internal session validation (19) before requesting Cart to process the order (20). Cart creates the Order (21), and Order creates an OrderLine for each cart line (22). TableMenu directly notifies the Employee of the new order (23) and confirms the order number, status, and total to the Customer (24). Bill and Payment are not included because they are outside the scope of UC-01.

5.4 UC-07 Pay Bill — scenario table and analysis sequence diagram

Trivial steps (no row): (0) the status page already holds the orders; (1) the customer reads the receipt. **Outside our system (no row):** between (8) and (9) the customer types card details on the gateway's page. **Non-trivial:** (1)–(6) build and show the bill; (7)–(9) confirm the amount and take it through the gateway; (10)–(12) record the payment and mark the orders paid; (13) show the receipt.

Step	Subject	Subject action	Other data / objects	Object acted upon
1	Customer	chooses to pay	—	TableMenu
2	TableMenu	asks for the unpaid orders of this table	table number	Order
3	TableMenu	asks for a bill to be built from those orders	unpaid orders	Bill
4	Bill	asks each order for its total	—	Order
5	Bill	adds the order totals into the amount due	order totals	Bill (itself)
6	TableMenu	shows the unpaid orders and the amount due	bill	Customer
7	Customer	confirms the amount	amount due	TableMenu

8	TableMenu	asks to take the amount for this bill	amount, bill reference	PaymentGateway
9	PaymentGateway	reports the payment approved	bill reference, gateway reference, amount	TableMenu
10	TableMenu	asks to record the payment on the bill	bill, gateway reference, amount	Bill
11	Bill	records an approved payment	amount, status (Approved), gateway reference, time	Payment
12	Bill	marks each order of the bill as paid by this payment	payment	Order
13	TableMenu	shows the receipt: amount, time, reference, orders	payment	Customer

UC-07 Pay Bill — sequence diagram

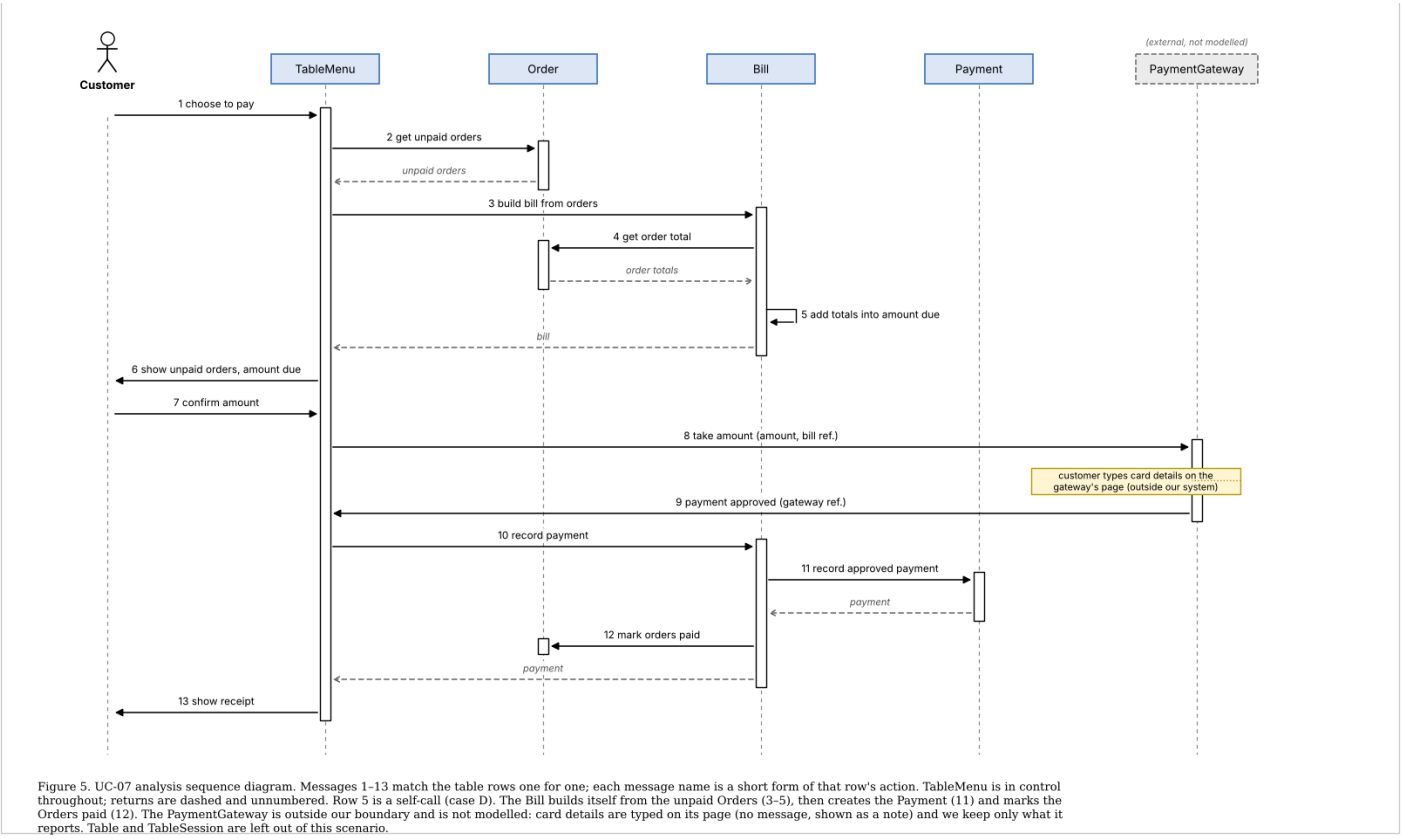


Figure 5. UC-07 analysis sequence diagram. Messages 1–13 match the table rows one for one; each message name is a short form of that row's action. TableMenu is in control throughout; returns are dashed and unnumbered. Row 5 is a self-call (case D). The Bill builds itself from the unpaid Orders (3–5), then creates the Payment (11) and marks the Orders paid (12). The PaymentGateway is outside our boundary and is not modelled: card details are typed on its page (no message, shown as a note) and we keep only what it reports. Table and TableSession are left out of this scenario.

Figure 5. UC-07 analysis sequence diagram. Messages 1–13 match the table rows; names are short forms of each row's action. The PaymentGateway is outside our boundary: we send it the amount (8) and it reports the result (9); its card handling is not modelled, and the customer types card details on its page. The Bill builds itself from the unpaid Orders (3–5), creates the Payment (11), and marks the Orders paid only at 12, after an approved result (FR-17).

5.5 Consistency cross-checks K1 to K6

Check	Rule [R7]	UC-01	UC-07
K1	Every non-trivial use-case step has a scenario row; trivial ones do not	Pass after D3	Pass
K2	Every item supplied or shown has a home in an object's responsibility	Pass after D1, D9	Pass after D4
K3	Every subject and receiver is in the object list, and every listed object is used	Pass after D5	Pass
K4	One row, one message; message numbers equal step numbers	Pass (19 = 19)	Pass (11 = 11)

Check	Rule [R7]	UC-01	UC-07
K5	Activation ends when the object finishes; returns dashed and unnumbered; no message above its cause	Pass after D6	Pass
K6	No technology in object or message names; outside services stay outside	Pass	Pass after D7

Defect log

#	Check	Who disagreed	What disagreed	Decision
D1	K2	Use case ↔ domain model	UC-01 (5) lets the customer type a note; no concept held it.	Added note to CartLine and OrderLine.
D2	K2	Use case ↔ domain model	UC-03 2a asks for a cancel reason; nowhere to keep it.	Added StatusChange.reason.
D3	K1	Use case ↔ table	First table had a row for step (4), which only shows data TableMenu already holds.	Step (4) marked trivial; row removed and rows renumbered.
D4	K2	Use case ↔ object list	The receipt shows a gateway reference; no inside object kept it.	Added gatewayReference to Payment.
D5	K3	Object list ↔ table	"StaffScreen" was listed but never acted upon.	Removed; StaffQueue is the receiver.
D6	K5	Table ↔ diagram	OrderBook's bar ended at 16, before it returned the order number.	Bar extended to the return after 17.
D7	K6	Object list ↔ diagram	First draft named the gateway "StripeCheckout" and drew a balance on it.	Renamed PaymentGateway, marked outside; no balance modelled.
D8	K3	Diagram ↔ P1 use case diagram	P1 diagram said payment would not be built, but FR-16/17 and UC-07 need it.	CR-03 replaced the P1 diagram.
D9	K2	Use case ↔ domain model	After the interview, UC-01 (2) offers a language and (4) shows allergens; neither had a home.	Added TableSession.language and MenuItem.allergens.
D10	K2	Use case ↔ domain model	UC-03 6a cancels one line, but cancellation lived only on Order.	Added OrderLine.cancelled and cancelReason.

6 High-Level Design

6.1 Quality-attribute scenarios

ID · quality	Stimulus and environment	Response	Response measure	From
QAS-1 · Performance	20 diners at 20 tables submit orders within the same minute during dinner rush; staff queue open on one tablet.	Each new order appears on every open staff screen without a reload.	≥ 95% of orders shown within 3 s of Submit (spike 01: 0.3 s at the 95th percentile).	QR-01, UC-01, UC-03
QAS-2 · Availability	The staff tablet loses its network for 60 s while 5 orders are submitted.	On reconnection the queue shows every order received meanwhile, in order.	0 orders missing; queue complete within 5 s of reconnection.	UC-03 1a, A2
QAS-3 · Security	Someone sends 1,000 requests to staff and manager functions without logging in, and 100 manager-only requests with a Staff login.	Every request is refused; nothing changes.	100% refused; 0 card numbers found in our data or logs by a full-text search.	QR-05, CON-02
QAS-4 · Modifiability	The restaurant changes to a different payment provider.	Only the part that talks to the provider changes.	1 component changed (C8); ≤ 3 person-days; no change to ordering.	UC-07, A5, Q1

6.2 Architectural style

Chosen: layered client–server (a single, modular application server) with publish–subscribe for live updates (ADR-01). Browsers run the customer and staff interfaces; one application server holds all rules in separate modules; one relational database stores the data. When an order changes, the order module publishes an event and a live-update module pushes it to every subscribed screen.

Style	QAS-1 speed	QAS-2 recovery	QAS-3 security	QAS-4 change	Cost for us	Verdict
Layered client–server, modular server + publish–subscribe	Push measured 0.3 s p95	Server keeps the queue; screen reloads it	One place to check every request	Payment isolated in one module	One deployable; fits 6 students and $50/month	**Chosen**
Microservices (menu, orders, payment as separate services)	Extra network hop per call	Each service must recover alone	Auth repeated in each service	Best isolation	Several deployables, a gateway, more hosting cost and ops work	Rejected
Serverless backend-as-a-service (client talks to a hosted database directly)	Good, built-in sync	Built-in	Rules written as database policies; easy to get wrong	Payment logic ends up in the client or functions	Vendor lock-in; prices and rules partly enforced on the phone	Rejected

6.3 Component view

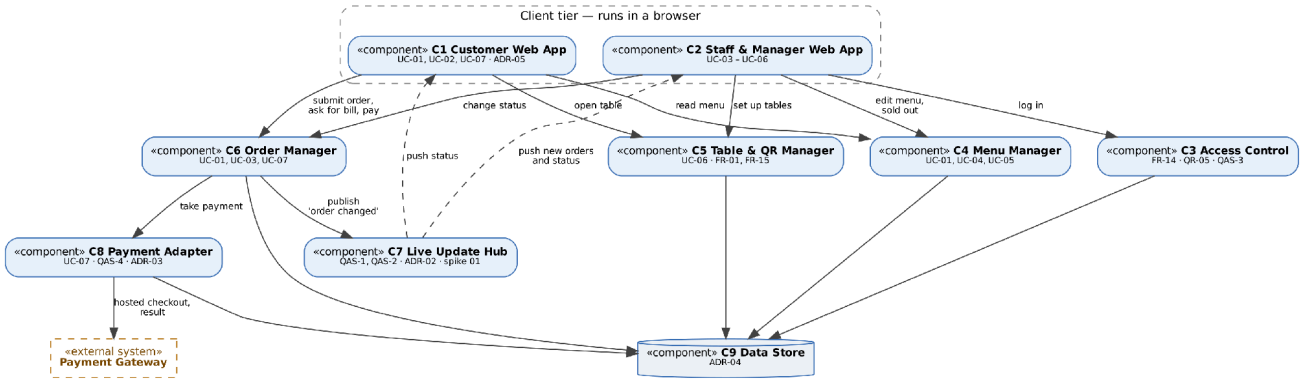


Figure 6. Component view. Each box names the use cases, scenarios or decisions it answers to. Solid arrows are requests; dashed arrows are pushes from the Live Update Hub.

Component	Responsibility	Answers to
C1 Customer Web App	Menu, cart, confirmation, order status and pay screens for one table.	UC-01, UC-02, UC-07; ADR-05
C2 Staff & Manager Web App	Staff queue, status changes, sold-out flags, menu and table editing.	UC-03 to UC-06
C3 Access Control	Logins, sessions and roles; checks every staff and manager request.	FR-14, QR-05, QAS-3
C4 Menu Manager	Categories, items, options, prices, availability.	UC-01, UC-04, UC-05
C5 Table & QR Manager	Tables, table links, QR images, active/inactive.	UC-06; FR-01, FR-15
C6 Order Manager	Shared table carts, orders, table checks, server confirmation, line cancellation, status changes, bills.	UC-01, UC-03, UC-07; FR-18, FR-21–FR-23, FR-27
C7 Live Update Hub	Pushes order events to subscribed screens; 1-s polling fallback.	QAS-1, QAS-2; ADR-02; spike 01
C8 Payment Adapter	Only component that talks to the payment gateway; verifies its signed results.	UC-07; QAS-4; ADR-03; CON-02
C9 Data Store	Keeps menu, tables, orders, bills, payments, staff accounts.	ADR-04

6.4 Deployment view

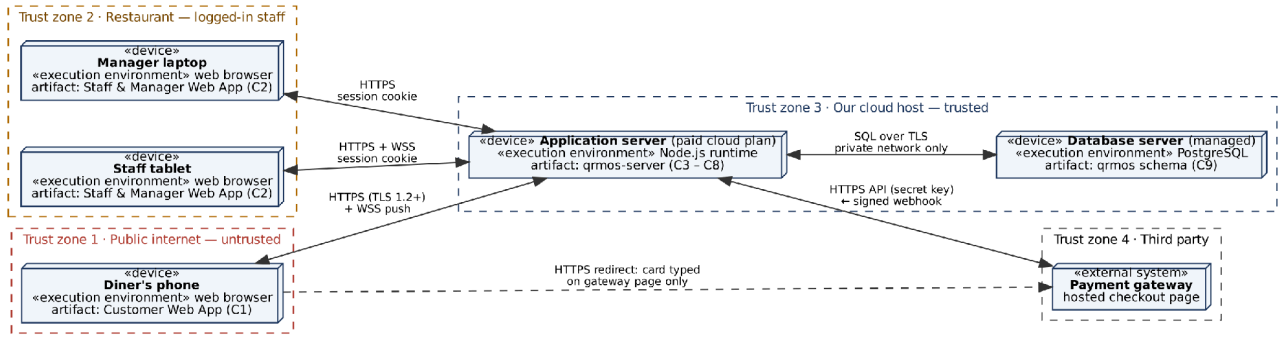


Figure 7. UML deployment view. Four trust zones (dashed). Every connection that crosses a zone is encrypted (HTTPS, WSS or TLS). The diner's phone talks to the gateway directly for card entry, so card data never crosses into zone 3.

Connection	Protocol	Boundary crossed	Protection
Phone ↔ application server	HTTPS + WSS	1 ⟶ 3	TLS; every request re-checked on the server; cart and prices held on the server (ADR-05)
Tablet / laptop ↔ application server	HTTPS + WSS	2 ⟶ 3	TLS; session cookie (HttpOnly, Secure); role check in C3
Application server ↔ database	SQL over TLS	inside 3	Private network only; no public address
Application server ↔ payment gateway	HTTPS API; signed webhook back	3 ↔ 4	Secret key kept in server configuration, never in the repository; webhook signature verified by C8
Phone ⟶ gateway checkout page	HTTPS redirect	1 ⟶ 4	Card data typed on the gateway's page only (CON-02)

6.5 Analysis-to-component mapping

Analysis object (5.2)	Component(s)
TableMenu	C1 Customer Web App (screens) + C6 Order Manager (rules it asks for)
Table	C5 Table & QR Manager
Menu, MenuItem	C4 Menu Manager
Cart	C6 Order Manager (one cart per table session, ADR-05) + C7 Live Update Hub (shows each phone's additions to the others)
OrderBook	C6 Order Manager
StaffQueue	C6 Order Manager (contents) + C7 Live Update Hub (delivery) + C2 (display)
Bill, Payment	C6 Order Manager
PaymentGateway	External; reached only through C8 Payment Adapter
StaffMember (UC-03 actor)	C3 Access Control

6.6 Design principles in this design

Principle	Where it shows
Separation of concerns	Three layers: browser apps (C1, C2), server modules (C3–C8), data (C9).
Information hiding	Only C8 knows the payment provider; swapping it touches one component (QAS-4).

Principle	Where it shows
High cohesion, low coupling	Each server module owns one group of domain concepts (6.5); modules call each other through narrow requests and publish events instead of sharing data.
Do not trust the client	Phones only send choices; C6 holds the shared cart and recomputes every price and availability on submit (ADR-05).
Least privilege	C3 checks the role on every staff and manager request; the database is reachable only from the server.
Single source of truth	An order exists only once C6 records it; screens show copies pushed by C7.

7 Design Rationale

7.1 ADR-01 Architectural style

Field	Content
Status · date	Accepted · 4 Nov 2026 (after CR-01 to CR-10)
Context	Six students, one term, $50/month hosting (2.6), QAS-1 to QAS-4.
Alternatives	Microservices; serverless backend-as-a-service; layered client–server with a modular server (6.2).
Decision	Layered client–server: browser apps, one modular application server, one relational database, publish–subscribe inside the server for live updates.
Cost	One deployable: a fault in any module can stop the whole server, and modules cannot be scaled separately. Accepted because one restaurant needs far less than one server can carry.
Linked to	QAS-1 to QAS-4; components C1–C9

7.2 ADR-02 Live order updates

Field	Content
Status · date	Accepted · 4 Nov 2026 · changed by spike 01
Context	QR-01 needs 95% of orders on the staff screen within 3 s; P1 risk R2 assumed 3-s polling would also do.
Alternatives	Polling only (every 3 s); server-sent events; WebSocket push with a polling fallback.
Decision	WebSocket push from C7 as the main channel; if the socket cannot open, the screen polls every 1 s.
Cost	The server keeps one open connection per screen and must handle reconnection (QAS-2). 1-s polling, when used, sends 3 times the requests of 3-s polling.
Linked to	QR-01, QAS-1, QAS-2, spike 01 (section 8), CR-07

7.3 ADR-03 Payment through a hosted checkout behind an adapter

Field	Content
Status · date	Accepted · 5 Nov 2026
Context	FR-16/17 need online payment; CON-02 forbids card data in our system; the interview named Square as the restaurant's provider, with the restaurant paying the fee (Q1).
Alternatives	Card form on our own page sent to the gateway's API; paying at the restaurant's till only; hosted checkout reached through one adapter.
Decision	Customers pay on the gateway's hosted page; only C8 talks to the gateway and verifies its signed result before C6 marks orders Paid.

Field	Content
Cost	Customers leave our page to pay, and we depend on the gateway's availability and its fee (about 3% per payment, A5). We cannot style the card form.
Linked to	UC-07, FR-16, FR-17, CON-02, QAS-3, QAS-4

7.4 ADR-04 Relational database

Field	Content
Status · date	Accepted · 5 Nov 2026
Context	Orders, lines, bills and payments are linked records that must change together (5.1).
Alternatives	Document store; relational database (PostgreSQL).
Decision	A managed PostgreSQL database; recording an order and its lines, or a payment and its Paid flags, happens in one transaction.
Cost	Schema changes need migrations; a managed plan is part of the $50/month hosting estimate.
Linked to	C9; FR-07, FR-17, FR-18

7.5 ADR-05 One shared cart per table, kept on the server

Field	Content
Status · date	Accepted · 8 Nov 2026 · supersedes the 6 Nov decision "cart kept on the phone"
Context	The manager interview asked that phones at one table share one cart (CR-17, FR-27). The 6 Nov decision kept each cart on its phone, which cannot be shared.
Alternatives	Cart on each phone, sent once on submit (the 6 Nov decision); cart on the server per table session, synced to every phone.
Decision	C6 keeps one cart per table session; every change is sent to the server and C7 pushes it to the other phones at the table. On submit C6 re-checks availability and prices (alternate 12a).
Cost	Every cart change is a request, so the menu needs a connection while ordering; two phones editing the same line at once need a rule (last change wins).
Linked to	UC-01, FR-04, FR-06, FR-18, FR-27; C6, C7; principle "do not trust the client"

7.6 Trade-off summary

Decision	We gain	We give up	We revisit if
ADR-01 layered, one server	Simple deployment, low cost, one place for rules	Independent scaling and fault isolation	More than one restaurant uses one install
ADR-02 push + 1-s polling	QR-01 met with margin (0.3 s p95)	Open connections to manage	P3 measurement on the real host misses QAS-1
ADR-03 hosted checkout	No card data, small PCI scope, provider swap in C8	Control of the payment page; fee per payment	The restaurant's provider has no hosted page (Q1)
ADR-04 relational DB	Order + lines and payment + Paid flags change together	Schema migrations	—
ADR-05 shared cart on server	One cart per table as the manager asked; cart survives a closed browser	More requests; ordering needs a connection	Spike re-run shows cart pushes slow QAS-1

8 Prototype Evidence

Question. Can a new order reach the staff screen within 3 seconds for 95% of orders while 20 tables order at once, and does polling every 3 seconds also meet that (as P1 risk R2 assumed)?

Spike. /spikes/spike-01-live-orders (owner Len Isherwood): a throw-away order server and a benchmark that sends 100 orders from 20 simulated tables (all 20 within 1 s of each other, five rounds) to one staff screen fed by push and one fed by polling, with 0, 80 and 150 ms of simulated one-way network delay. Three runs per setting; raw results in results.jsonl [R8].

One-way delay	Push p95	Push within 3 s	Poll 3 s p95	Poll 3 s within 3 s	Poll 1 s p95	Poll 1 s within 3 s
0 ms	5 ms	100%	2,705 ms	100%	—	—
80 ms	164 ms	100%	3,037 ms	93%	—	—
150 ms	304 ms	100%	3,558 ms	44%	1,517 ms	100%

Result. Push meets QR-01 with a wide margin; 3-second polling does not once real network delay is added (93% and 44% within 3 s, below the 95% target); 1-second polling does.

What it changed. ADR-02 keeps push (confirmed) and changes the fallback from 3-s to 1-s polling (changed). Risk R2 and its mitigation were corrected (CR-07).

Limits. Measured on one machine with simulated delay, without a database in the path, on 29 Oct 2026. P3 repeats the measurement on the deployed host with the real database (10.2).

9 Traceability

Requirement	Use case	Analysis objects	Component(s)	Scenario / decision
FR-01 QR per table (MUST)	UC-06, UC-01	Table	C5	—
FR-02 menu, no login (MUST)	UC-01	TableMenu, Menu	C1, C4	ADR-01
FR-03 item details (MUST)	UC-01	Menu, MenuItem	C1, C4	—
FR-04 no sold-out items	UC-01, UC-04	Menu	C4, C6	ADR-05
FR-05 options, note (MUST)	UC-01	MenuItem, Cart	C1, C4	—
FR-06 edit cart	UC-01	Cart	C1	ADR-05
FR-07 submit with table (MUST)	UC-01	TableMenu, OrderBook	C6, C9	ADR-04
FR-08 confirmation	UC-01	OrderBook	C6, C1	—
FR-09 live queue (MUST)	UC-03	StaffQueue	C6, C7, C2	QAS-1, ADR-02
FR-10 status changes (MUST)	UC-03	OrderBook	C6	QAS-2
FR-11 customer status	UC-02	—	C7, C1	ADR-02
FR-12 edit menu (MUST)	UC-05	—	C4, C2	—
FR-13 mark sold out	UC-04	—	C4, C2	—
FR-14 staff login (MUST)	Log In «include»	—	C3	QAS-3
FR-15 tables	UC-06, UC-01 2a	Table	C5	—
FR-16 pay online	UC-07	TableMenu, OrderBook, Bill, PaymentGateway	C1, C6, C8	ADR-03, QAS-4
FR-17 Paid only if approved	UC-07	OrderBook, Payment, Bill	C6, C8	ADR-03
FR-18 no duplicate order	UC-01 13a	OrderBook	C6, C9	ADR-04

Requirement	Use case	Analysis objects	Component(s)	Scenario / decision
FR-19 time since received	UC-03	StaffQueue	C2, C6	—
FR-20 receipt	UC-07	Payment	C1, C6	—
FR-21 table check	UC-03, UC-07	StaffQueue, OrderBook, Bill	C6, C2	—
FR-22 cancel one line	UC-03 6a	OrderBook	C6, C2	—
FR-23 server confirms	UC-03	StaffQueue, OrderBook	C6, C2, C7	QAS-1
FR-24 English/Spanish	UC-01	Menu, MenuItem	C4, C1	—
FR-25 allergens	UC-01	MenuItem	C4, C1	—
FR-26 payment-failure alert	UC-07 6a	Payment	C8, C7, C2	ADR-03
FR-27 shared cart	UC-01	Cart	C6, C7	ADR-05
QR-01 / QR-02 speed	UC-01, UC-03	—	C7, C1	QAS-1, ADR-02, spike 01
QR-05 security · CON-02 card data	all staff UCs · UC-07	—	C3 · C8	QAS-3, ADR-03

Gap analysis.

1. **FR-11, FR-12, FR-13 have no analysis objects**, because UC-02, UC-04 and UC-05 were left at the high level (4.1). They still reach a component. Action: expand UC-02 in P3 if FR-11 testing shows the status view needs its own objects.
2. **QR-03 (usability) and QR-04 (accessibility) reach no component decision**; they are tested on C1 in P3 (usability test, axe and screen-reader checks).
3. **QR-06 (availability) has no scenario**: it is COULD priority and depends on the host's uptime. Recorded, not acted on.
4. **C7 now also carries shared-cart updates (FR-27) and payment alerts (FR-26)**, beyond the order pushes the spike measured. P3 re-runs the spike with cart traffic added (10.2).
5. **FR-23 adds a human step before the kitchen.** QR-01 measures delivery to the staff screen, which still holds; the time a server takes to confirm is outside the system and is not measured.
6. **Every MUST requirement reaches at least one component**, and every requirement realised by UC-01, UC-03 and UC-07 is traced through analysis objects to a component.

10 Project Plan

10.1 Team roles and working agreement

Team member	Role for P2 to P3	How the team decides when it disagrees
Mohmmad Abuzetoon	Project manager, repository owner, release and tagging	Discuss in the weekly meeting or team chat; majority vote after 24 hours; the section owner breaks a tie; every decision is an issue labelled "decision".
Rafay Alvi	Requirements and change control; acceptance tests from 3.1	Same rule (/docs/team-agreement.md)
Eshitasri Punuganti	Use cases, traceability; P3 test cases from section 4	Same rule
Xuan Mai Tran	Detailed design (Chapter 9 class and sequence diagrams)	Same rule
Isaiah Bueno-Hernandez	Architecture owner; server modules C3–C8	Same rule
Len Isherwood	Live updates (C7) and performance re-measurement	Same rule

10.2 Plan to P3

Date	What will be finished	Owner
8 Nov	P2 committed to /docs/p2, tag p2 pushed, PDF uploaded to Canvas	Mohmmad Abuzetoon
14 Nov	Follow-up with a server and a kitchen staff member (the interview reached only the manager); Q7 funding answer	Isaiah Bueno-Hernandez
12 Nov	Detailed design: design class diagram and design sequence diagrams for UC-01 and UC-07	Xuan Mai Tran
15 Nov	Server skeleton for C3–C6 with the database schema (ADR-04)	Isaiah Bueno-Hernandez
19 Nov	Live Update Hub (C7) on the deployed host; spike 01 re-measured with the real database and shared-cart traffic	Len Isherwood
22 Nov	Customer and staff web apps for UC-01 and UC-03 end to end	Rafay Alvi, Eshitasri Punuganti
26 Nov	Payment adapter (C8) against the gateway's test mode; UC-07 end to end	Isaiah Bueno-Hernandez
29 Nov	Test plan and acceptance tests from 3.1 and 3.2 (Chapter 11)	Eshitasri Punuganti, Rafay Alvi
P3 deadline (Canvas)	P3 document, code and test results committed and tagged	Mohmmad Abuzetoon

10.3 Risks

#	Risk	Likelihood	Impact	Mitigation — action and owner
R1	The seven interview changes (CR-11 to CR-17) add work before P3, and the server/kitchen follow-up may add more.	Medium	High	Isaiah Bueno-Hernandez raises any change as a CR before 12 Nov; Rafay Alvi re-runs 3.5 and section 9.
R2	Push updates fail on the chosen host (connections closed by a proxy).	Medium	High	Len Isherwood re-runs spike 01 on the host by 19 Nov; fallback is 1-s polling, measured to meet QR-01 (CR-07).
R3	Scope creep: split bills, POS integration, loyalty.	Medium	Medium	Rafay Alvi checks every new idea against 1.2 and change control (0.2).
R4	A member stops contributing.	Medium	High	Mohmmad Abuzetoon reviews the issue board every Monday, reassigns work overdue by 3 days, and tells the instructor at once.
R5	Labour runs over 600 hours, pushing cost toward the ceiling.	Low	Medium	Mohmmad Abuzetoon logs hours weekly from 9 Nov; past 450 hours by 1 Dec, SHOULD and COULD items are dropped first (CR-10).
R6	The payment gateway's test mode is unavailable or needs paperwork.	Medium	Medium	Isaiah Bueno-Hernandez opens a test account by 15 Nov; if blocked, C8 is tested against a recorded gateway response.

10.4 Repository and tooling

Repository: https://github.com/mrzetoon/team-01-QRRestaurantOrder — public (or instructor read access before the deadline).

Folders: /docs/p1 (P1-submitted.pdf, the file on Canvas) · /docs/p2 (this PDF and its .docx) · /docs/p2/diagrams (every diagram's source: .dot, .py, and the images) · /docs/elicitation (interview notes) · /docs/meetings (weekly minutes) · /docs/ai-log.md · /spikes/spike-01-live-orders · /src (from P3).

Branches and pull requests: main is protected; each member works on p2/<section> and merges through a pull request reviewed by another member.

Issues: one per P2 section and one per change request, labelled doc, requirement, spike or decision, with an owner and a due date.

Release: the P2 PDF is committed to /docs/p2 and tagged p2 (git tag p2; git push origin p2) before the same file is uploaded to Canvas by Mohmmad Abuzetoon. No passwords or API keys are committed; the gateway key lives in the host's configuration.

10.5 Declaration of AI tool use

Tool	What it was used for	Who used it	What we checked or changed
Claude (Anthropic)	P1: first draft of requirements, use cases and diagrams.	Mohmmad Abuzetoon	Requirements re-checked for testability; stakeholder sources rewritten; placeholders removed.
Claude (Anthropic)	P2: draft wording of sections 0.2 and 4–9, diagram scripts (/docs/p2/diagrams), and the spike 01 code.	Mohmmad Abuzetoon	Each owner read and corrected their section; Xuan Mai Tran re-ran K1–K6; Len Isherwood re-ran the spike and checked results.jsonl.

11 Open Questions

#	The question	Who can answer it	Needed by	Status · what we assume meanwhile
Q1	Which payment gateway does the restaurant use, and who pays its fee?	Restaurant owner (S4)	8 Nov	Answered [R9]: Square; the restaurant pays the processing fee. ADR-03 uses Square's hosted checkout.
Q2	Is a second order from a table a new order or part of the first?	Manager, server	8 Nov	Answered [R9]: added to the table's original ticket. CR-11 (FR-21 table check).
Q3	Can the kitchen cancel one item instead of the whole order?	Kitchen staff (S3)	8 Nov	Answered [R9]: yes, single items. CR-12 (FR-22).
Q4	Must a server confirm each order before the kitchen sees it?	Manager (S4)	8 Nov	Answered [R9]: yes, to avoid mistakes. CR-13 (FR-23).
Q5	Which languages must the menu support?	Manager (S4)	8 Nov	Answered [R9]: English and Spanish. CR-14 (FR-24).
Q6	Must allergen information have a fixed format?	Manager (S4)	8 Nov	Answered [R9]: common allergens listed on the menu; staff answer questions. CR-15 (FR-25).
Q7	Will the restaurant fund about $43,300 to build and $4,200 a year to run?	Restaurant owner (S4)	14 Nov	Open (not asked at the interview) · the 2.6 estimate stands.
Q8	If a payment fails or cannot be confirmed, what should staff see?	Manager (S4)	8 Nov	Answered [R9]: a clear payment-failure notification so staff can help the customer pay again. CR-16 (FR-26).
Q9	Should two phones at one table share one cart?	Manager, server	8 Nov	Answered [R9]: yes. CR-17 (FR-27); ADR-05 reversed.
Q10	When two phones change the same cart line at once, which change wins?	Manager (S4)	14 Nov	New (ADR-05) · the last change wins.
Q11	Should drinks or simple items skip server confirmation?	Manager, server	14 Nov	New (FR-23) · every order is confirmed.

12 Baseline Declaration

We confirm that this document, version 1.0, is our agreed requirements and design baseline as of the submission date, that the PDF on Canvas is the file tagged p2 in the repository, and that every team member has read it in full.

Team member	Signature (typed name)	Date
Mohmmad Abuzetoon	Mohmmad Abuzetoon	8 Nov 2026
Rafay Alvi	Rafay Alvi	8 Nov 2026
Eshitasri Punuganti	Eshitasri Punuganti	8 Nov 2026

Team member	Signature (typed name)	Date
Xuan Mai Tran	Xuan Mai Tran	8 Nov 2026
Len Isherwood		
Isaiah Bueno-Hernandez	Isaiah Bueno-Hernandez	8 Nov 2026